

## DAVENPORT, IA, USA

WMO: 744550

Lat: 41.614N

Lon: 90.591W

Elev: 750

StdP: 14.30

Time Zone: -6.00 (NAC)

Period: 05-19

WBAN: 94982

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -7.1       | -0.8 | -13.5                         | 2.7 | -6.4 | -8.7 | 3.5 | -0.4 | 30.3                  | 26.6 | 27.5 | 21.7 | 8.6                   | 300  | 0.562 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 19.3                   | 90.8            | 74.9 | 88.4 | 74.1 | 86.1 | 72.8 | 78.0                | 87.5 | 76.5 | 85.5 | 74.9 | 83.3 | 11.1                 | 200  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 75.2                            | 135.8 | 84.1 | 73.4 | 127.5 | 82.6 | 71.9 | 121.4 | 80.8 | 42.0          | 87.3 | 40.3 | 85.8 | 38.8 | 83.3 | 83.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |       |    |    |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|-------|----|----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |       |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max   | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)   |    |    |
| 26.6              | 23.9 | 20.5 | -15.3                      | 94.6 | 6.8                | 2.6 | -20.2                                              | 96.5 | -24.2      | 98.0 | -28.0      | 99.4 | -32.9      | 101.2 |    |    |
|                   |      |      | -15.8                      | 81.2 | 6.5                | 1.3 | -20.5                                              | 82.2 | -24.4      | 82.9 | -28.0      | 83.7 | -32.8      | 84.6  |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 50.2   | 22.1  | 24.6  | 38.9  | 50.5  | 62.5  | 71.5  | 73.9  | 71.8  | 65.7  | 52.6  | 39.3  | 27.8  |
|                                                                                  | DBStd       | 20.84  | 12.73 | 12.73 | 11.85 | 9.78  | 9.03  | 5.79  | 5.85  | 5.12  | 8.17  | 10.19 | 10.46 | 11.72 |
|                                                                                  | HDD50       | 3212   | 866   | 714   | 382   | 111   | 11    | 0     | 0     | 0     | 2     | 90    | 344   | 692   |
|                                                                                  | HDD65       | 6315   | 1331  | 1131  | 811   | 442   | 157   | 11    | 5     | 6     | 90    | 406   | 771   | 1154  |
|                                                                                  | CDD50       | 3295   | 0     | 2     | 39    | 127   | 399   | 646   | 740   | 675   | 472   | 170   | 23    | 2     |
|                                                                                  | CDD65       | 926    | 0     | 0     | 3     | 7     | 81    | 207   | 280   | 216   | 111   | 21    | 0     | 0     |
|                                                                                  | CDH74       | 8563   | 0     | 0     | 23    | 120   | 748   | 1897  | 2738  | 1827  | 1015  | 195   | 0     | 0     |
|                                                                                  | CDH80       | 2746   | 0     | 0     | 1     | 15    | 211   | 620   | 996   | 534   | 325   | 44    | 0     | 0     |
| Wind                                                                             | WSAvg       | 9.9    | 11.5  | 11.4  | 11.6  | 12.3  | 10.8  | 8.8   | 6.9   | 6.5   | 7.6   | 9.4   | 10.9  | 10.7  |
| Precipitation                                                                    | PrecAvg     | 37.5   | 1.3   | 1.5   | 2.5   | 3.7   | 4.7   | 5.2   | 4.2   | 4.3   | 3.0   | 2.6   | 2.2   | 1.8   |
|                                                                                  | PrecMax     | 51.8   | 2.6   | 3.8   | 5.6   | 7.0   | 9.9   | 14.0  | 11.3  | 9.6   | 8.7   | 6.9   | 6.1   | 3.8   |
|                                                                                  | PrecMin     | 19.2   | 0.4   | 0.1   | 0.5   | 1.3   | 0.1   | 1.0   | 0.4   | 1.1   | 0.8   | 0.5   | 0.3   | 0.4   |
|                                                                                  | PrecStd     | 7.5    | 0.7   | 1.0   | 1.7   | 1.5   | 2.2   | 3.1   | 2.6   | 2.2   | 1.9   | 1.6   | 1.5   | 0.9   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 53.6  | 63.1  | 77.3  | 82.2  | 90.6  | 91.5  | 95.2  | 91.4  | 90.9  | 85.3  | 70.3  | 59.5  |
|                                                                                  |             | MCWB   | 44.8  | 53.4  | 62.6  | 61.8  | 71.2  | 74.3  | 78.5  | 75.2  | 74.0  | 70.6  | 57.1  | 54.9  |
|                                                                                  | 2%          | DB     | 47.6  | 53.4  | 69.2  | 77.9  | 85.5  | 89.6  | 91.4  | 88.2  | 87.5  | 79.6  | 64.2  | 53.0  |
|                                                                                  |             | MCWB   | 43.1  | 47.1  | 57.8  | 61.1  | 68.5  | 74.1  | 76.5  | 74.9  | 71.8  | 65.5  | 54.9  | 47.8  |
|                                                                                  | 5%          | DB     | 42.2  | 46.5  | 63.0  | 72.5  | 81.7  | 86.6  | 88.2  | 85.7  | 83.8  | 74.4  | 59.8  | 47.7  |
|                                                                                  |             | MCWB   | 38.6  | 41.4  | 54.5  | 58.4  | 66.0  | 72.1  | 75.4  | 73.9  | 69.8  | 62.6  | 51.7  | 44.2  |
|                                                                                  | 10%         | DB     | 37.9  | 40.9  | 56.4  | 68.1  | 78.1  | 83.5  | 85.5  | 82.9  | 80.2  | 69.5  | 55.2  | 43.0  |
|                                                                                  |             | MCWB   | 35.1  | 37.3  | 49.0  | 55.2  | 64.4  | 71.0  | 73.7  | 71.9  | 67.6  | 59.8  | 48.8  | 39.3  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 48.7  | 55.2  | 64.2  | 65.5  | 73.5  | 78.3  | 81.0  | 79.0  | 76.1  | 72.4  | 60.3  | 56.6  |
|                                                                                  |             | MCDB   | 51.3  | 59.5  | 75.3  | 75.9  | 86.4  | 88.3  | 89.7  | 87.6  | 87.1  | 82.8  | 65.5  | 59.7  |
|                                                                                  | 2%          | WB     | 42.8  | 47.6  | 59.9  | 63.0  | 70.9  | 76.4  | 78.6  | 77.1  | 74.2  | 68.9  | 56.4  | 48.4  |
|                                                                                  |             | MCDB   | 47.1  | 53.1  | 67.3  | 74.1  | 81.8  | 86.1  | 88.2  | 85.8  | 83.4  | 76.2  | 62.6  | 51.1  |
|                                                                                  | 5%          | WB     | 38.7  | 42.2  | 55.1  | 60.1  | 68.7  | 74.5  | 77.1  | 75.3  | 72.2  | 65.1  | 53.0  | 44.1  |
|                                                                                  |             | MCDB   | 42.0  | 46.5  | 61.6  | 70.6  | 78.1  | 83.6  | 86.3  | 82.9  | 80.2  | 71.2  | 59.3  | 47.5  |
|                                                                                  | 10%         | WB     | 35.3  | 37.3  | 49.6  | 57.1  | 66.5  | 72.6  | 75.4  | 73.5  | 70.2  | 60.8  | 48.9  | 39.1  |
|                                                                                  |             | MCDB   | 37.7  | 40.8  | 56.4  | 66.2  | 75.4  | 81.3  | 83.6  | 80.4  | 77.8  | 68.4  | 54.8  | 42.5  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 15.2  | 16.6  | 18.2  | 21.4  | 20.6  | 19.3  | 19.3  | 19.5  | 22.4  | 20.7  | 18.1  | 14.0  |
|                                                                                  |             | MCDBR  | 18.6  | 23.5  | 25.3  | 28.6  | 24.0  | 21.6  | 20.6  | 20.8  | 23.6  | 25.5  | 23.6  | 18.7  |
|                                                                                  |             | MCWBR  | 15.1  | 17.4  | 16.2  | 16.0  | 11.7  | 10.6  | 9.8   | 10.5  | 10.9  | 13.6  | 15.8  | 15.0  |
|                                                                                  | 5% WB       | MCDBR  | 17.6  | 22.0  | 22.3  | 24.2  | 21.0  | 19.1  | 18.3  | 17.9  | 19.7  | 19.9  | 21.9  | 18.0  |
|                                                                                  |             | MCWBR  | 14.6  | 17.1  | 15.2  | 15.0  | 11.4  | 10.6  | 10.0  | 10.0  | 10.2  | 13.0  | 16.2  | 15.6  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.292  | 0.302 | 0.342 | 0.386 | 0.419 | 0.423 | 0.441 | 0.421 | 0.392 | 0.344 | 0.302 | 0.290 |       |
|                                                                                  | taud        | 2.403  | 2.391 | 2.373 | 2.259 | 2.231 | 2.270 | 2.205 | 2.263 | 2.315 | 2.449 | 2.502 | 2.435 |       |
|                                                                                  | Ebn at Noon | 273    | 288   | 287   | 280   | 272   | 271   | 264   | 266   | 266   | 269   | 269   | 264   |       |
|                                                                                  | Edh at Noon | 25     | 30    | 34    | 42    | 44    | 42    | 45    | 41    | 36    | 28    | 23    | 23    |       |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 559    | 872   | 1186  | 1481  | 1736  | 1947  | 1933  | 1708  | 1446  | 951   | 632   | 462   |       |
|                                                                                  | RadStd      | 66     | 90    | 100   | 128   | 125   | 171   | 128   | 113   | 119   | 110   | 53    | 48    |       |

## Historical Trends

|      |                         | DBAvg | Heating |        |       | Cooling |       |     | Degree-Days |       |       |       |
|------|-------------------------|-------|---------|--------|-------|---------|-------|-----|-------------|-------|-------|-------|
|      |                         |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |     | HDD50       | HDD65 | CDD50 | CDD65 |
|      |                         | (d)   | (e)     | (f)    | (g)   | (h)     | (i)   | (j) | (k)         | (l)   | (m)   |       |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   |       |
| (47) | Regional (48 neighbors) | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   |       |

Nomenclature: See separate page